

4.4.3 Internet Control Message Protocol (ICMP)

Recall that the network layer of the Internet has three main components: the IP protocol, discussed in the previous section; the Internet routing protocols (including RIP, OSPF, and BGP), which are covered in Section 4.6; and ICMP, which is the subject of this section.

ICMP, specified in [RFC 792], is used by hosts and routers to communicate network-layer information to each other. The most typical use of ICMP is for error reporting. For example, when running a Telnet, FTP, or HTTP session, you may have encountered an error message such as “Destination network unreachable.” This message had its origins in ICMP. At some point, an IP router was unable to find a path to the host specified in your Telnet, FTP, or HTTP application. That router created and sent a type-3 ICMP message to your host indicating the error.

ICMP is often considered part of IP but architecturally it lies just above IP, as ICMP messages are carried inside IP datagrams. That is, ICMP messages are carried as IP payload, just as TCP or UDP segments are carried as IP payload. Similarly, when a host receives an IP datagram with ICMP specified as the upper-layer protocol, it demultiplexes the datagram’s contents to ICMP, just as it would demultiplex a datagram’s content to TCP or UDP.

ICMP messages have a type and a code field, and contain the header and the first 8 bytes of the IP datagram that caused the ICMP message to be generated in the first place (so that the sender can determine the datagram that caused the error). Selected ICMP message types are shown in Figure 4.23. Note that ICMP messages are used not only for signaling error conditions.

The well-known ping program sends an ICMP type 8 code 0 message to the specified host. The destination host, seeing the echo request, sends back a type 0 code 0 ICMP echo reply. Most TCP/IP implementations support the ping server directly in the operating system; that is, the server is not a process. Chapter 11 of [Stevens 1990] provides the source code for the ping client program. Note that the client program needs to be able to instruct the operating system to generate an ICMP message of type 8 code 0.

Another interesting ICMP message is the source quench message. This message is seldom used in practice. Its original purpose was to perform congestion control—to allow a congested router to send an ICMP source quench message to a host to force that host to reduce its transmission rate. We have seen in Chapter 3 that TCP has its own congestion-control mechanism that operates at the transport layer, without the use of network-layer feedback such as the ICMP source quench message.

In Chapter 1 we introduced the Traceroute program, which allows us to trace a route from a host to any other host in the world. Interestingly, Traceroute is implemented with ICMP messages. To determine the names and addresses of the routers between source and destination, Traceroute in the source sends a series of ordinary IP datagrams to the destination. Each of these datagrams carries a UDP segment with an unlikely UDP port number. The first of these datagrams has a TTL of 1, the

ICMP Type	Code	Description
0	0	echo reply (to ping)
3	0	destination network unreachable
3	1	destination host unreachable
3	2	destination protocol unreachable
3	3	destination port unreachable
3	6	destination network unknown
3	7	destination host unknown
4	0	source quench (congestion control)
8	0	echo request
9	0	router advertisement
10	0	router discovery
11	0	TTL expired
12	0	IP header bad

Figure 4.23 ♦ ICMP message types

second of 2, the third of 3, and so on. The source also starts timers for each of the datagrams. When the n th datagram arrives at the n th router, the n th router observes that the TTL of the datagram has just expired. According to the rules of the IP protocol, the router discards the datagram and sends an ICMP warning message to the source (type 11 code 0). This warning message includes the name of the router and its IP address. When this ICMP message arrives back at the source, the source obtains the round-trip time from the timer and the name and IP address of the n th router from the ICMP message.

How does a Traceroute source know when to stop sending UDP segments? Recall that the source increments the TTL field for each datagram it sends. Thus, one of the datagrams will eventually make it all the way to the destination host. Because this datagram contains a UDP segment with an unlikely port number, the destination host sends a port unreachable ICMP message (type 3 code 3) back to the source. When the source host receives this particular ICMP message, it knows it does not need to send additional probe packets. (The standard Traceroute program actually sends sets of three packets with the same TTL; thus the Traceroute output provides three results for each TTL.)



FOCUS ON SECURITY

INSPECTING DATAGRAMS: FIREWALLS AND INTRUSION DETECTION SYSTEMS

Suppose you are assigned the task of administering a home, departmental, university, or corporate network. Attackers, knowing the IP address range of your network, can easily send IP datagrams to addresses in your range. These datagrams can do all kinds of devious things, including mapping your network with ping sweeps and port scans, crashing vulnerable hosts with malformed packets, flooding servers with a deluge of ICMP packets, and infecting hosts by including malware in the packets. As the network administrator, what are you going to do about all those bad guys out there, each capable of sending malicious packets into your network? Two popular defense mechanisms to malicious packet attacks are firewalls and intrusion detection systems (IDSs).

As a network administrator, you may first try installing a firewall between your network and the Internet. (Most access routers today have firewall capability.) Firewalls inspect the datagram and segment header fields, denying suspicious datagrams entry into the internal network. For example, a firewall may be configured to block all ICMP echo request packets, thereby preventing an attacker from doing a traditional ping sweep across your IP address range. Firewalls can also block packets based on source and destination IP addresses and port numbers. Additionally, firewalls can be configured to track TCP connections, granting entry only to datagrams that belong to approved connections.

Additional protection can be provided with an IDS. An IDS, typically situated at the network boundary, performs “deep packet inspection,” examining not only header fields but also the payloads in the datagram (including application-layer data). An IDS has a database of packet signatures that are known to be part of attacks. This database is automatically updated as new attacks are discovered. As packets pass through the IDS, the IDS attempts to match header fields and payloads to the signatures in its signature database. If such a match is found, an alert is created. An intrusion prevention system (IPS) is similar to an IDS, except that it actually blocks packets in addition to creating alerts. In Chapter 8, we’ll explore firewalls and IDSs in more detail.

Can firewalls and IDSs fully shield your network from all attacks? The answer is clearly no, as attackers continually find new attacks for which signatures are not yet available. But firewalls and traditional signature-based IDSs are useful in protecting your network from known attacks.

In this manner, the source host learns the number and the identities of routers that lie between it and the destination host and the round-trip time between the two hosts. Note that the Traceroute client program must be able to instruct the operating system to generate UDP datagrams with specific TTL values and must also be able to be notified by its operating system when ICMP messages arrive. Now that you understand how Traceroute works, you may want to go back and play with it some more.

4.4.4 IPv6

In the early 1990s, the Internet Engineering Task Force began an effort to develop a successor to the IPv4 protocol. A prime motivation for this effort was the realization that the 32-bit IP address space was beginning to be used up, with new subnets and IP nodes being attached to the Internet (and being allocated unique IP addresses) at a breathtaking rate. To respond to this need for a large IP address space, a new IP protocol, IPv6, was developed. The designers of IPv6 also took this opportunity to tweak and augment other aspects of IPv4, based on the accumulated operational experience with IPv4.

The point in time when IPv4 addresses would be completely allocated (and hence no new networks could attach to the Internet) was the subject of considerable debate. The estimates of the two leaders of the IETF's Address Lifetime Expectations working group were that addresses would become exhausted in 2008 and 2018, respectively [Solensky 1996]. In February 2011, IANA allocated out the last remaining pool of unassigned IPv4 addresses to a regional registry. While these registries still have available IPv4 addresses within their pool, once these addresses are exhausted, there are no more available address blocks that can be allocated from a central pool [Huston 2011a]. Although the mid-1990s estimates of IPv4 address depletion suggested that a considerable amount of time might be left until the IPv4 address space was exhausted, it was realized that considerable time would be needed to deploy a new technology on such an extensive scale, and so the Next Generation IP (IPng) effort [Bradner 1996; RFC 1752] was begun. The result of this effort was the specification of IP version 6 (IPv6) [RFC 2460] which we'll discuss below. (An often-asked question is what happened to IPv5? It was initially envisioned that the ST-2 protocol would become IPv5, but ST-2 was later dropped.) Excellent sources of information about IPv6 are [Huitema 1998, IPv6 2012].

IPv6 Datagram Format

The format of the IPv6 datagram is shown in Figure 4.24. The most important changes introduced in IPv6 are evident in the datagram format:

- *Expanded addressing capabilities.* IPv6 increases the size of the IP address from 32 to 128 bits. This ensures that the world won't run out of IP addresses. Now, every grain of sand on the planet can be IP-addressable. In addition to unicast and multicast addresses, IPv6 has introduced a new type of address, called an **anycast address**, which allows a datagram to be delivered to any one of a group of hosts. (This feature could be used, for example, to send an HTTP GET to the nearest of a number of mirror sites that contain a given document.)
- *A streamlined 40-byte header.* As discussed below, a number of IPv4 fields have been dropped or made optional. The resulting 40-byte fixed-length header allows